VALLABH PATIL

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SUMMARY

AI Engineer with hands-on experience building and improving LLM-based systems, including RAG pipelines and AI agents. I've worked on evaluating model outputs, identifying failure cases, and improving response quality through prompt design and retrieval strategies. Comfortable working across Python-based ML systems and backend services, with a focus on building practical, scalable solutions. Particularly interested in model evaluation, reliability, and how AI systems behave in real-world use.

EDUCATION

MSc in Advanced Computer Science — **Cardiff, UK**
Cardiff University | Grade : **Distinction |** Received Best Dissertation (Specialist Degree) Award — **September 2023 - September 2024**

PG-Diploma in Artificial Intelligence — **Pune, India**
Centre for Development of Advanced Computing | Grade : **A** — **March 2023 - August 2023**

Bachelor of Engineering in Computer Science — **Pune, India**
SKN College of Engineering | Grade : **7.77 / 10 CGPA** — **August 2016 - May 2020**

SKILLS & INTERESTS

Machine Learning & LLMs: LLMs, RAG, Prompt Engineering, LangChain, LlamaIndex, NLP
Model Evaluation: Output analysis, error analysis, performance metrics, response quality improvement
Programming : Python, FastAPI, Django
Data & ML Tools: PyTorch, TensorFlow, Hugging Face, Scikit-learn
MLOps & Systems: Docker, AWS (EC2, S3), REST APIs
Databases: PostgreSQL, MySQL, Vector DBs (Weaviate, Chroma, Qdrant)

PROFESSIONAL EXPERIENCE

Full Stack Developer and Automation Engineer, N&S Consultants — **October 2025 - Present**
Birmingham, UK
- Led development of an AI-powered platform for medico-legal expert search, reducing manual CV matching time by ~60–70%.
- Built a RAG-based system to retrieve and rank expert profiles, improving relevance and reducing manual filtering effort.
- Developed backend services using Python (FastAPI) with PostgreSQL and Redis, improving response times by ~30%.
- Implemented automated expert recommendation and quote generation, reducing processing time by ~40%.
- Evaluated model outputs to identify failure cases and improve retrieval quality through prompt and search optimisation.
- Designed and deployed the system end-to-end, including architecture, frontend (React), and monitoring with Prometheus.

AI Backend Service Engineer, Risidio — **January 2025 - May 2025**
London, UK
- Worked on LLM-based backend services, including AI agents and RAG pipelines, using Python and tools such as LangChain and LlamaIndex.
- Designed and improved retrieval pipelines, leading to more relevant responses and a 20% reduction in query latency.
- Evaluated model outputs across different query types, identifying common failure cases and improving response quality through prompt adjustments and retrieval tuning.
- Reduced token usage and operational cost by optimising prompt structures and implementing caching strategies for repeated queries.
- Built and maintained APIs using FastAPI and integrated vector databases (Weaviate) for efficient semantic search.
- Collaborated within an Agile team, contributing to sprint planning, code reviews, and system design discussions.

Associate Software Engineer, Accenture — **January 2021 - July 2022**
Bangalore, India
- Worked on backend systems using Python, supporting feature updates, bug fixes, and production deployments.
- Improved database query performance, resulting in a 15% increase in efficiency.
- Managed multiple change requests and releases, ensuring stable and reliable deployments.
- Collaborated in Agile teams, participating in sprint planning, testing, and delivery of new features.

PROJECTS

Log Classification System Using Hybrid Classification (NLP) | Github Link — **2025**
- Built a hybrid log classification system combining rule-based methods, machine learning, and LLMs to handle varying log complexity.
- Evaluated model performance using a labelled dataset, improving classification accuracy by 40% compared to regex-only approaches.
- Analysed misclassifications and refined feature selection and model behaviour to improve reliability.
- Developed a FastAPI service to expose the system for real-time usage.

Document RAG Application | Github Link — **2025**
- Developed a RAG-based application for document querying and summarisation using LLMs.
- Evaluated response relevance and summarisation quality across multiple documents, achieving up to 85% accuracy.
- Improved retrieval performance and response consistency by optimising embeddings and vector search strategies.
- Built a scalable backend using FastAPI and integrated session-based vector storage.

Ontology-Driven Medical Data Analysis (Master's Dissertation) — **2024**
- Conducted ontology-driven analysis of medical data using **NLP** techniques and transfer learning with **BERT** and **BioBERT** LLMs.
- Utilized **Named Entity Recognition (NER)** for extracting key medical entities from PubMed articles.
- Leveraged OCD ontology for robust feature extraction and classification tasks.
- Demonstrated proficiency in **Python, BERT/BioBert LLMs, TensorFlow, Scikit-learn, NER** and Owlready for ontology integration and data processing.

CERTIFICATIONS AND AWARDS

- Received Best MSc Dissertation (Specialist Degree) annual graduation award, Cardiff University — **July 2025**
- Building Generative AI with AWS: Amazon Q Developer, Bedrock Inference, and SageMaker Canvas, LinkedIn — **Feb 2025**
- Generative AI Fundamentals, Databricks — **Feb 2025**
- Build a Backend REST API with Python & Django - Advanced, Udemy — **July 2024**